

Week 1 - Core Data Analytics Foundation Report

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Data Analytics Internship

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Objective

The objective of this project is to learn the fundamentals of Data Analytics by performing data cleaning, SQL analysis, Python-based exploratory data analysis, and creating interactive dashboards using Excel and Power BI.

Dataset

Dataset Name:

Sample Superstore Dataset

Source:

Kaggle / Tableau Sample Superstore Dataset

Records:

9694

Columns:

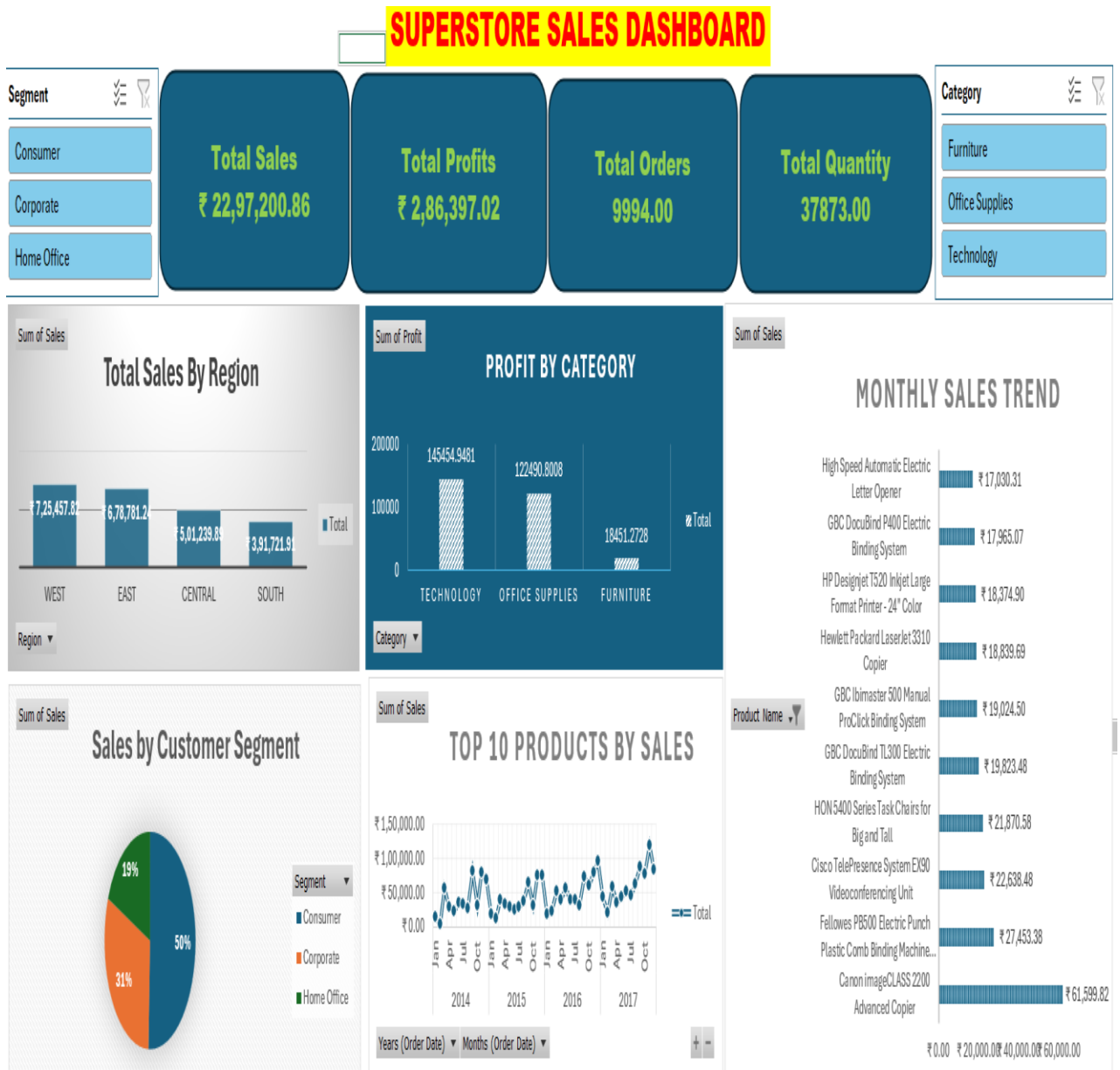
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Tools Used

- Microsoft Excel
- MySQL Workbench
- Google Colab (Python)
- Power BI Desktop

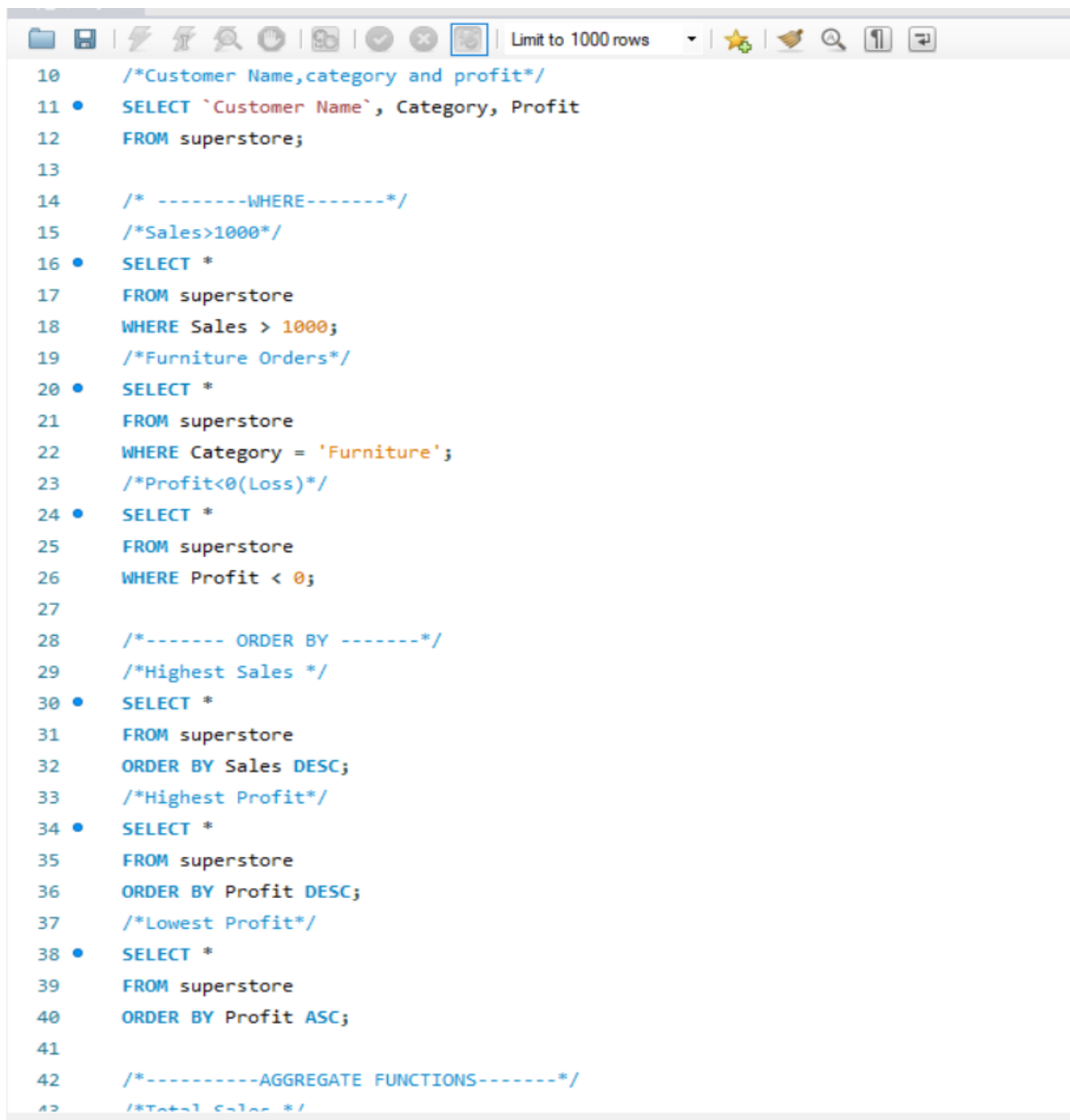
Excel Work

- Data Cleaning
- Date Formatting
- Removing Missing Values
- Pivot Tables
- KPI Calculations
- Interactive Excel Dashboard



SQL Analysis

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- Aggregate Functions
- HAVING
- Subqueries



```
10      /*Customer Name,category and profit*/
11 •    SELECT `Customer Name`, Category, Profit
12      FROM superstore;
13
14      /* -----WHERE-----*/
15      /*Sales>1000*/
16 •    SELECT *
17      FROM superstore
18      WHERE Sales > 1000;
19      /*Furniture Orders*/
20 •    SELECT *
21      FROM superstore
22      WHERE Category = 'Furniture';
23      /*Profit<0(Loss)*/
24 •    SELECT *
25      FROM superstore
26      WHERE Profit < 0;
27
28      /*----- ORDER BY -----*/
29      /*Highest Sales */
30 •    SELECT *
31      FROM superstore
32      ORDER BY Sales DESC;
33      /*Highest Profit*/
34 •    SELECT *
35      FROM superstore
36      ORDER BY Profit DESC;
37      /*Lowest Profit*/
38 •    SELECT *
39      FROM superstore
40      ORDER BY Profit ASC;
41
42      /*-----AGGREGATE FUNCTIONS-----*/
43      /*Total Sales */
```

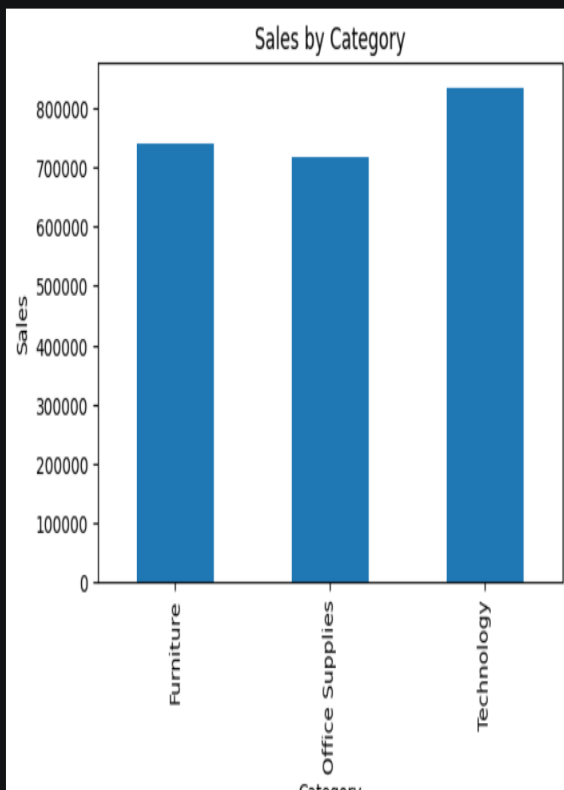
Python Analysis

- Pandas
- NumPy
- Matplotlib
- Data Loading
- Summary Statistics
- Missing Value Analysis
- Sales & Profit Analysis

Sales by Category

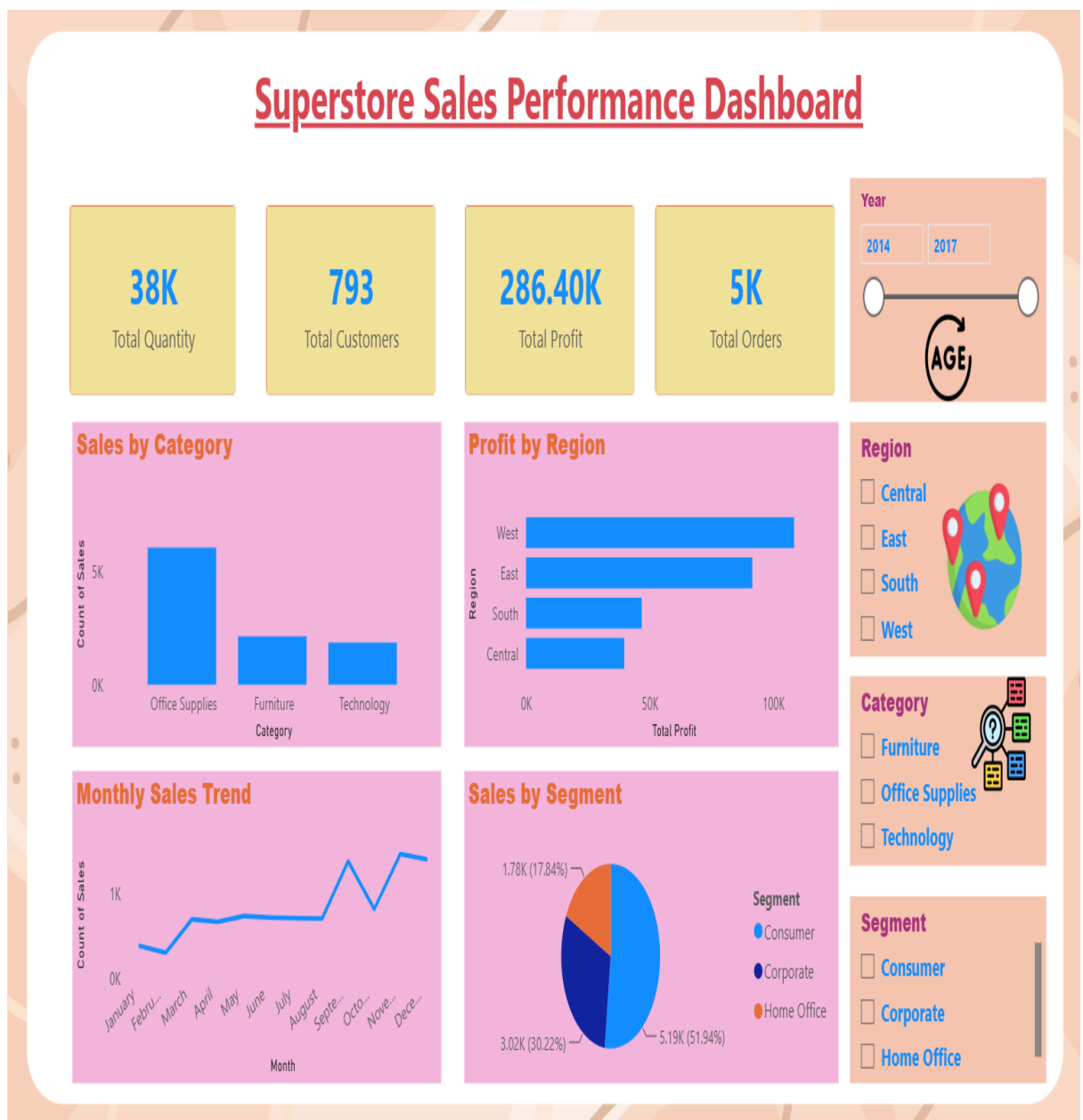
```
sales_category = df.groupby("Category")["Sales"].sum()

plt.figure(figsize=(6,4))
sales_category.plot(kind="bar")
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```



Power BI Dashboard

- KPI Cards
- Sales by Category
- Sales by Region
- Sales by Segment
- Monthly Sales Trend
- Top Products
- Interactive Slicers



Key Insights

- Technology category generated the highest sales.
 - West region recorded the highest revenue.
 - Consumer segment contributed the largest share of sales.
 - Monthly sales showed an overall increasing trend.
 - A small number of products contributed significantly to total sales.
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Conclusion

This project strengthened my understanding of Excel, SQL, Python, and Power BI. I learned how to clean data, perform exploratory data analysis, write SQL queries, create dashboards, and generate business insights from real-world datasets.